

Landmark-Focused vs Turn-by-Turn Navigation in Minecraft Virtual Environments

An Exploratory Investigation of Interface

Design, Task Complexity, and Spatial Knowledge Acquisition

Andrea Bragante

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University of Copenhagen



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1 Introduction

- **Spatial knowledge and spatial navigation**
- Navigation aids
- Interfaces
- Why Minecraft?
- Research question and hypotheses

Spatial knowledge:

- Informations about space, its locations, relationships between places.
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- *survey* knowledge: map-like environment representation

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Spatial navigation

The process of planning a route and moving in space towards a destination.

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- eg. paper maps, in-car navigation systems

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Different aids impact differently, eg.:

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- **GPS-based** interfaces, such as Google Maps, are efficient [1]...

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- **GPS-based** interfaces, such as Google Maps, are efficient [1]... but also associated with
 - less active spatial processing [4]
 - spatial memory impairment [5]
 - lower accuracy when forming cognitive maps [6]

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Interface design can help: role of **landmarks** [7], [8]

1 Introduction

- Spatial knowledge and spatial navigation
- Navigation aids
- **Interfaces**
- Why Minecraft?
- Research question and hypotheses

Interfaces

- **Landmark-focused (LF)**
 - only shows landmark names
 - encourage use of landmarks
- **Turn-by-turn (TBT)**
 - familiar, commercial model
 - directional cues (e.g. go straight, turn left)
 - guiding line



Figure 1: LF on the top, TBT on the bottom

1 Introduction

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- **Why Minecraft?**
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Minecraft: open-ended, cube-based videogame.

- advantages in using videogames
 - intuitive commands
 - reproducibility
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- advantages in using videogames
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 - eg. *Xaero's minimap* → our interfaces

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- Interfaces
- Why Minecraft?
- **Research question and hypotheses**

Research Question

How different is the **impact** of *landmark-focused (LF)* and *turn-by-turn (TBT)* interfaces on *survey and route knowledge* in a **navigation task** within a **Minecraft** virtual environment?

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H1 (spatial knowledge and navigation)

- **LF** yields (compared to TBT)
 - more accurate survey knowledge
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How different is the **impact** of *landmark-focused (LF)* and *turn-by-turn (TBT)* interfaces on *survey and route knowledge* in a **navigation task** within a **Minecraft** virtual environment?

H1 (spatial knowledge and navigation)

- **LF** yields (compared to TBT)
 - more accurate survey knowledge
 - shorter return travel time

H2 (eye-tracking, exploratory)

- TBT and LF produce different gaze patterns.
- **LF** will look (compared to TBT)
 - less at the interface
 - more at the environment

2 Methods

- **Experiment in a nutshell**
- Independent variables: interface and task
- Dependent variables: performance and eye-tracking
- Statistical analyses

Experiment design

- 16 participants
- 4 navigation tasks

Task phases

- **learning** phase, navigation **ON**, towards goal
- **recall** phase, navigation **OFF**, back to starting point

Individual skills **questionnaires**

- Santa-Barbara Sense of Direction (SBSOD)
- Perspective Taking/Spatial Orientation Test (PTSOT)
- Experience with Videogames and Minecraft, scale 1-3

2 Methods

- Experiment in a nutshell
- **Independent variables: interface and task**
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2.1 Independent variables: interface and task



Interface

- **LF** only shows landmark names
- **TBT** also shows navigational cues (on intersections) and guiding line



Figure 2: LF on the top, TBT on the bottom

2.1 Independent variables: interface and task



Task

Four tasks with different difficulty.

Factors used:

- number of landmarks
- number of turns
- total path length

Difficulty scale

Task 2 >> Task 1 and 4 >> Task 3



Figure 3: Minimap top-view of the four tasks

2 Methods

- Experiment in a nutshell
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Performance variables

- Route knowledge
 - Learning phase travel time (*seconds*)
 - Recall phase *extra* travel time (*seconds*)

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 - Landmark-placement error (*pixels*)
 - Pointing error (*degrees*)
 - Pointing decision time (*seconds*)

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Eye-tracking variables

- Fixation duration
- Fixation number

2 Methods

- Experiment in a nutshell
- Independent variables: interface and task
- Dependent variables: performance and eye-tracking
- **Statistical analyses**

Analysis structure

- Python and R for data processing and analysis.
- **Linear Mixed-Effects Models (LMMs)**
 - fixed effects for task and interface
 - random effects for individual differences
- **Wald's tests** to test main effects of
 - Interface
 - Task
 - Interaction *Interface* × *Task*

3 Results and Discussion

- **Performance variables**
- Eye-tracking variables

Task effect

Significant across most variables
→ Task design was effective.

Interface effect

- **No** effect on survey knowledge
- Effect on route knowledge
 - Interaction in recall extra time

Performance variables	Task	Interface	Interface×Task
route: Learning Time	* * * *	* * *	$p > 0.5$
route: Recall Extra Time	* * * *	* *	* *
survey: Landmark pl. error	* * *	$p > 0.5$	$p > 0.5$
survey: Pointing error	* *	$p > 0.5$	$p > 0.5$
survey: Pointing decision time	-	$p > 0.5$	-

Table 1: Results of Wald's tests, only p-values (page 33)

*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$

3.1 Performance variables



Learning phase - travel time

Interface effect

- TBT significantly faster than LF

TBT interface design was *effective*.

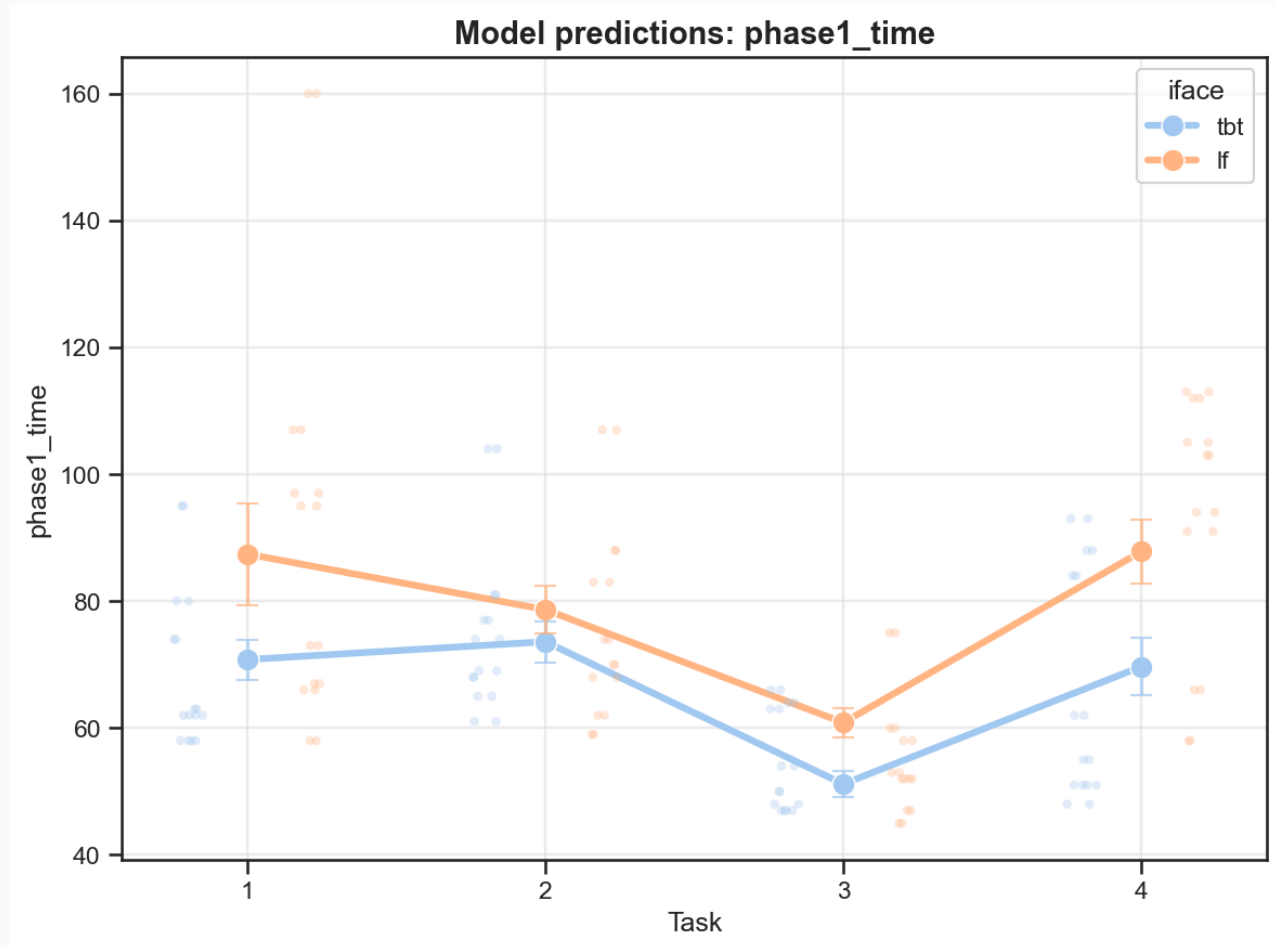


Figure 4: Learning travel time predictions, page 34

3.1 Performance variables



Recall phase - travel time

- *Crossover pattern*: effect depends on the interaction *Interface* × *Task*
- Interface effect
 - LF > TBT in Tasks 1 and 2
 - LF ≥ in Tasks 3 and 4

Interface *matters*; task difficulty can change effect direction.

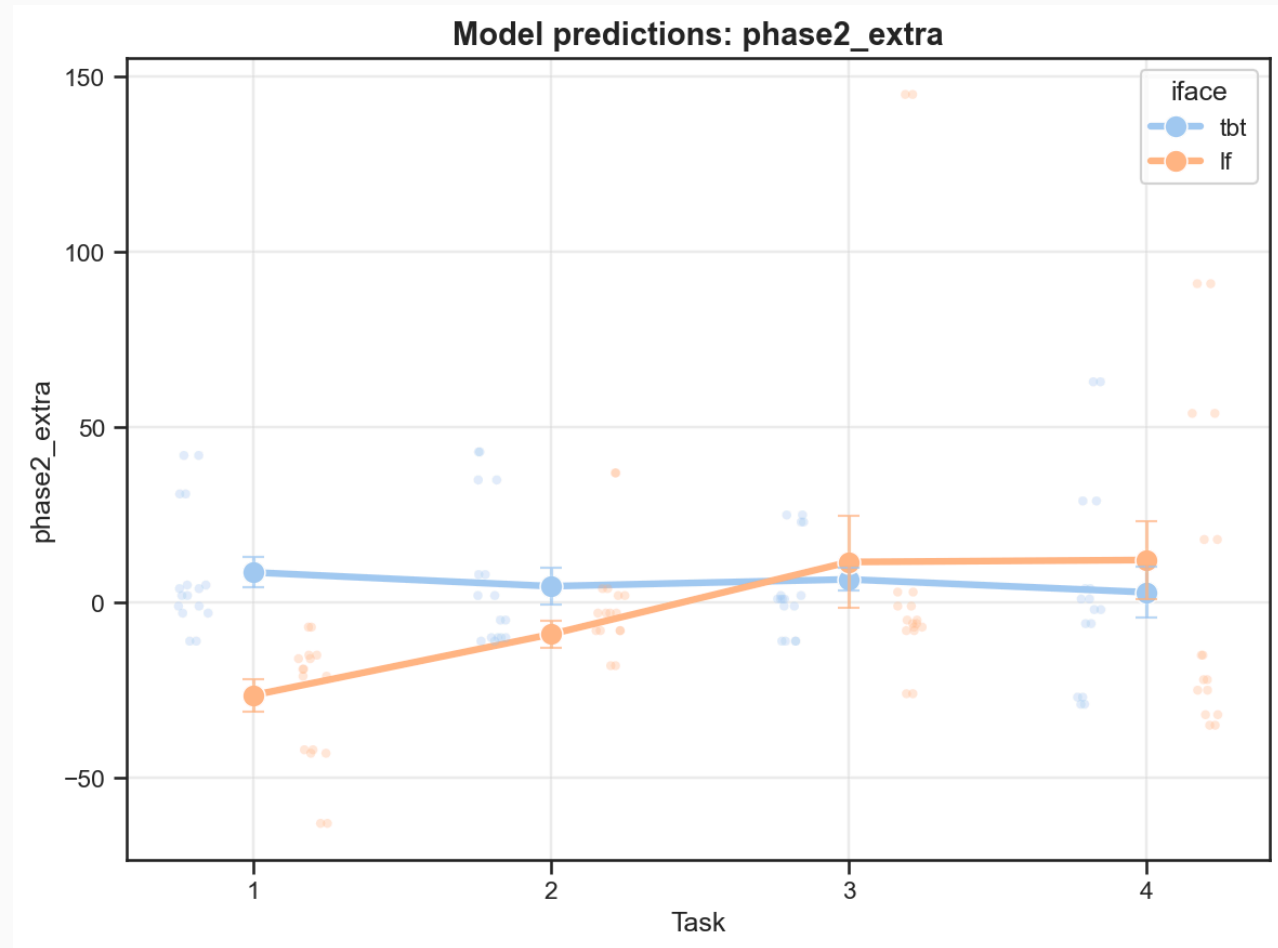


Figure 5: Recall travel time predictions, page 35

Landmark placement error

- Task difficulty has significant effect
- Interface effect **not** significant
 - Task 1 (TBT: -56px)
 - Task 2* (TBT: +85px)

Task effect dominates.

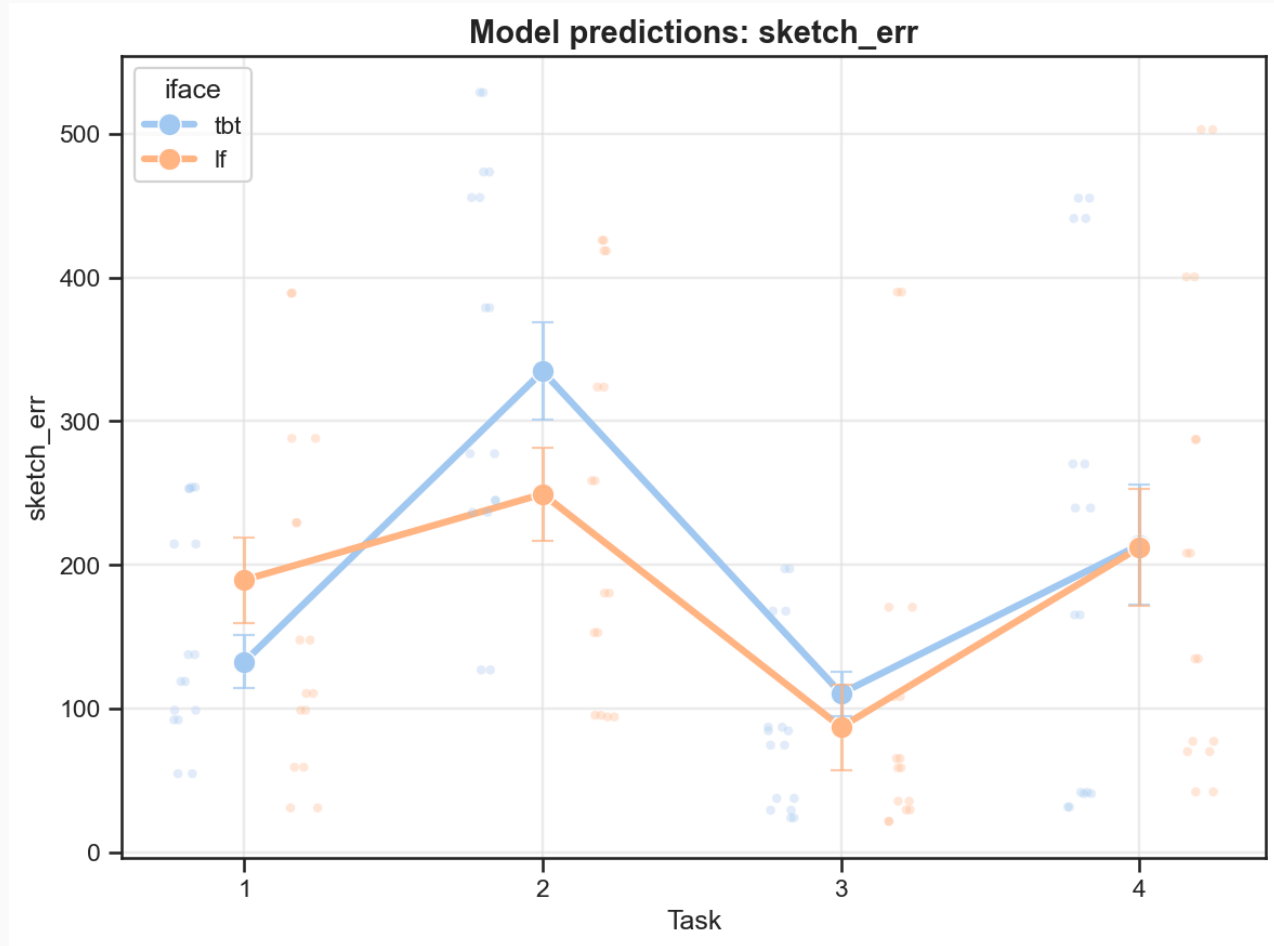


Figure 6: Landmark placement error predictions, page 36

3.1 Performance variables



Pointing angle error

- Task difficulty has significant effect
- Interface effect **not** significant
 - Task 3 (TBT: -9°)

Task effect dominates.



Figure 7: Angle error predictions, page 37

3.1 Performance variables



Pointing decision time

Not informative.

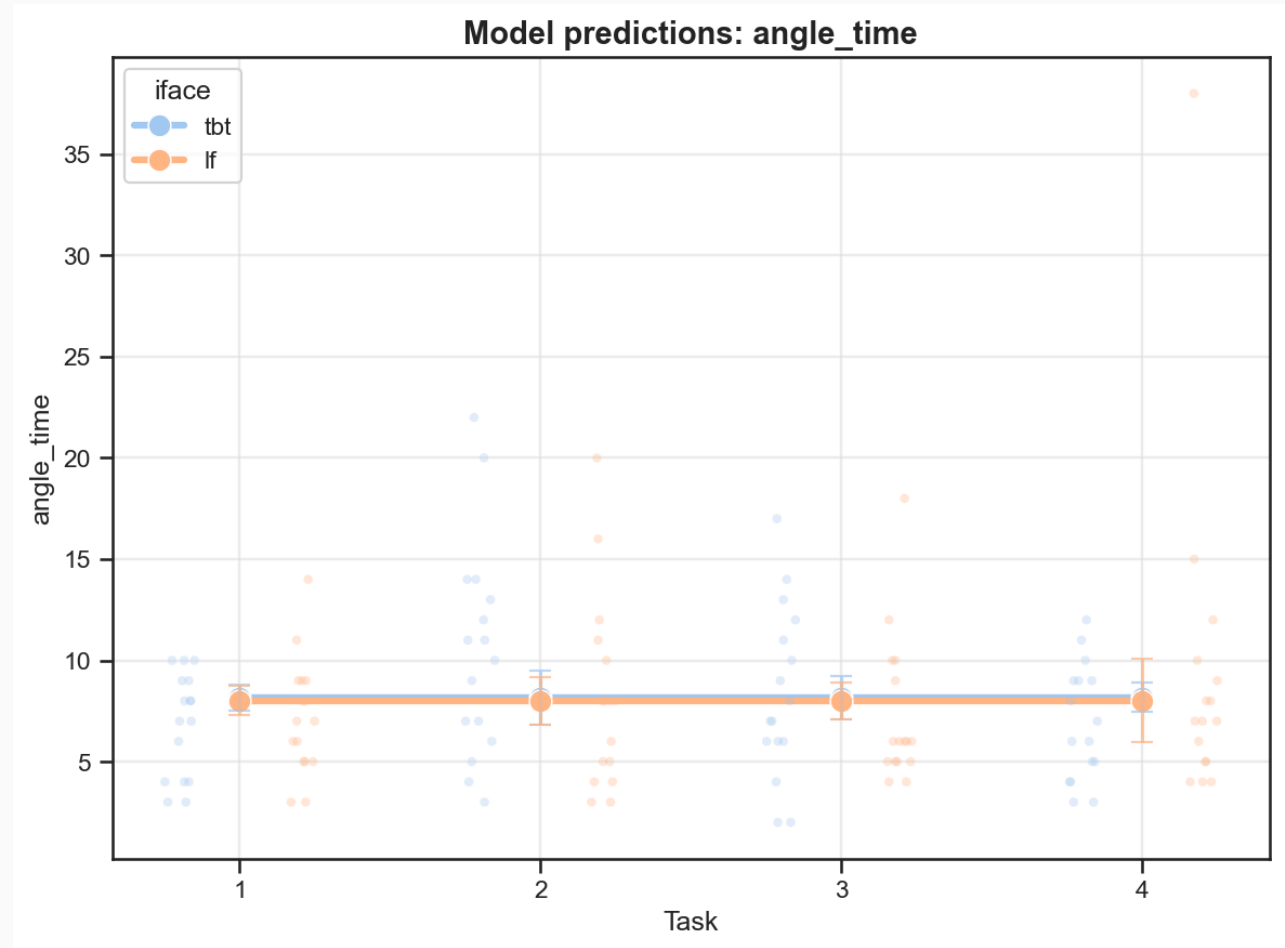


Figure 8: Angle time predictions, page 37

3 Results and Discussion

- Performance variables
- **Eye-tracking variables**

Number of fixations

- Interface effect is **not** significant overall
 - Significantly interacts with AOI

Area of Interest (AOI): environment, nav. cues, minimap

Effect	pvalue
Intercept	****
interface	$p > 0.1$
task	****
aoi	****
interface:task	*
interface:aoi	****
task:aoi	****

Table 2: Main effects and interactions between interface, task, and AOI, only p-values, page 41

3.1 Eye-tracking variables



Number of fixations - AOI x Interface

AOI	LF	TBT	LF/TBT	p-value
Env	84.8	74.0	1.15	****
Minimap	59.9	44.3	1.35	****

Table 3: AOI x Interface interaction, page 41

LF fixates

- 15% more on the environment
- 35% more on the interface

compared to TBT.

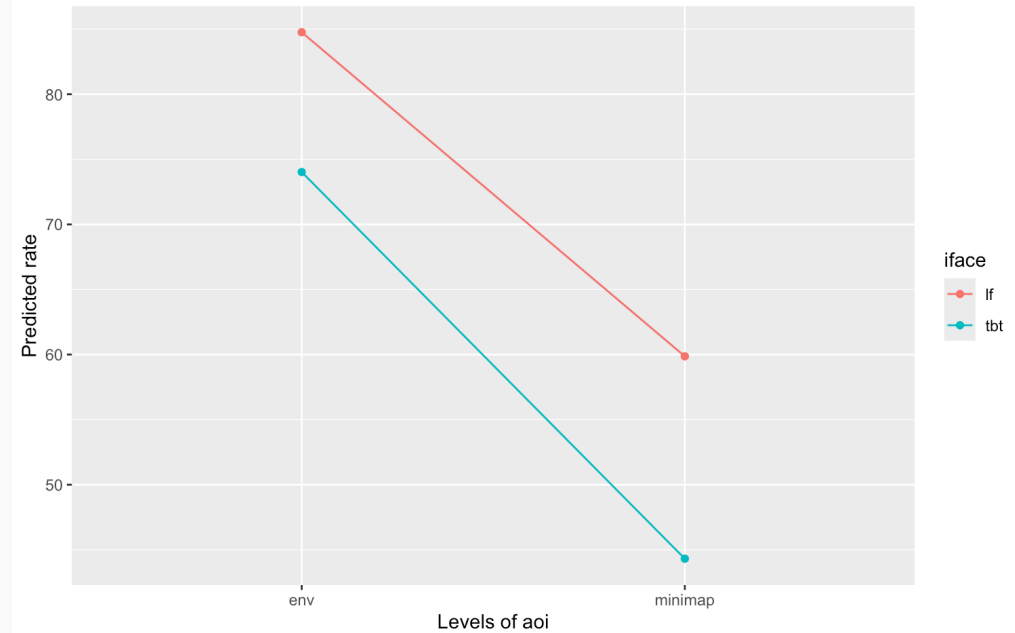


Figure 9: AOI x Interface interaction, page 41

3.1 Eye-tracking variables



Ratio of fixations

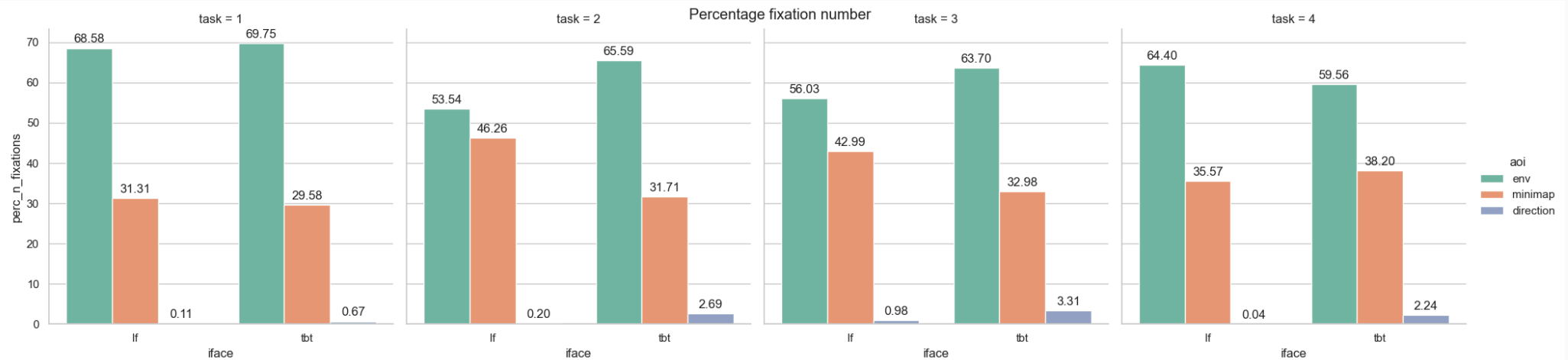


Figure 10: Ratio of fixations, page 40

TBT directional cues are ignored

4 Conclusions

- Interface effect
- Task effect
- Gaze

Interface can affect navigation performance (H1 part. rejected)

- TBT improves navigation efficiency (learning phase)
- LF *can be* good for *recall* phase, depending on the task
- No significant impact on survey measures (survey knowledge acquired later [2])

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Task effect dominance

- Task difficulty: stronger predictor of performance than interface type.
- Difficulty rankings confirmed post-hoc

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Task effect dominance

- Task difficulty: stronger predictor of performance than interface type.
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Gaze effect (H2 rejected)

- LF fixates more, especially on the interface

Thanks!

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Model selection for performance variables

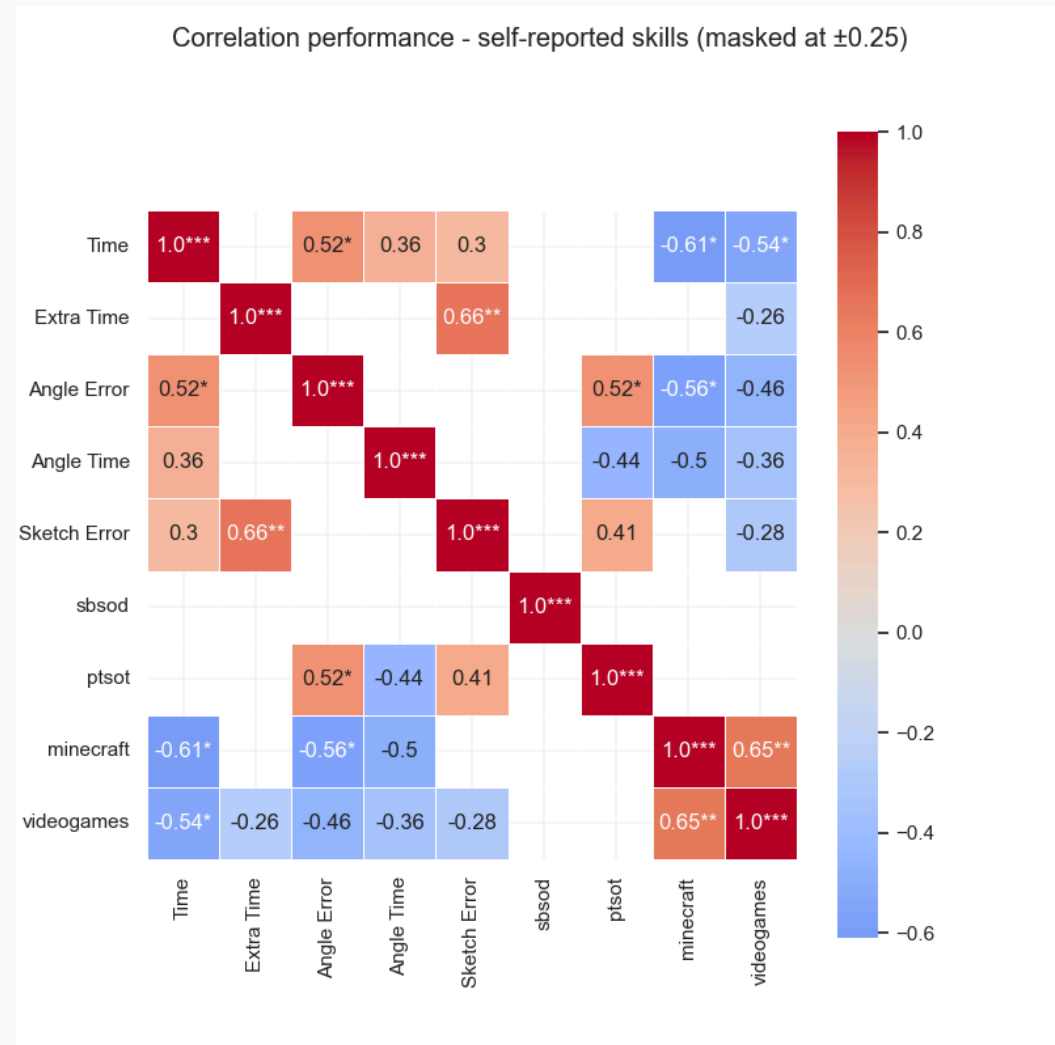
- M4 for times: interface effect varies across participants (individual differences)
- M3 for sketch and angle error ...
- M1 for angle time ...

<i>Dependent variable</i>	<i>Summary</i>
Learning phase time	M4: Interface x Task, Rand. Slope on Interface
Recall phase time	M4: Interface x Task, Rand. Slope on Interface
Landmark-placement error	M3: Interface x Task
Angle time	M1: Interface effect only
Angle error	M3: Interface x Task

Table 4: Summary of model selection process results, showing the selected model for each dependent variable. The formula syntax follows the one in *lme4* package for R.

Correlations

- travel time associated with pointing error and sketch error
- SBSOD no correlations
- PTSOT: angle error (0.52); likely poor spatial orientation skills harm performance, consistent with results from the literature (Hejtmánek et al., 2018)
- Minecraft and Videogames positively impact performance



- what are the conclusions to draw from the results?
- what are the problems in which we incurred?
- how would i redesign the experiment? how would i go next from here?